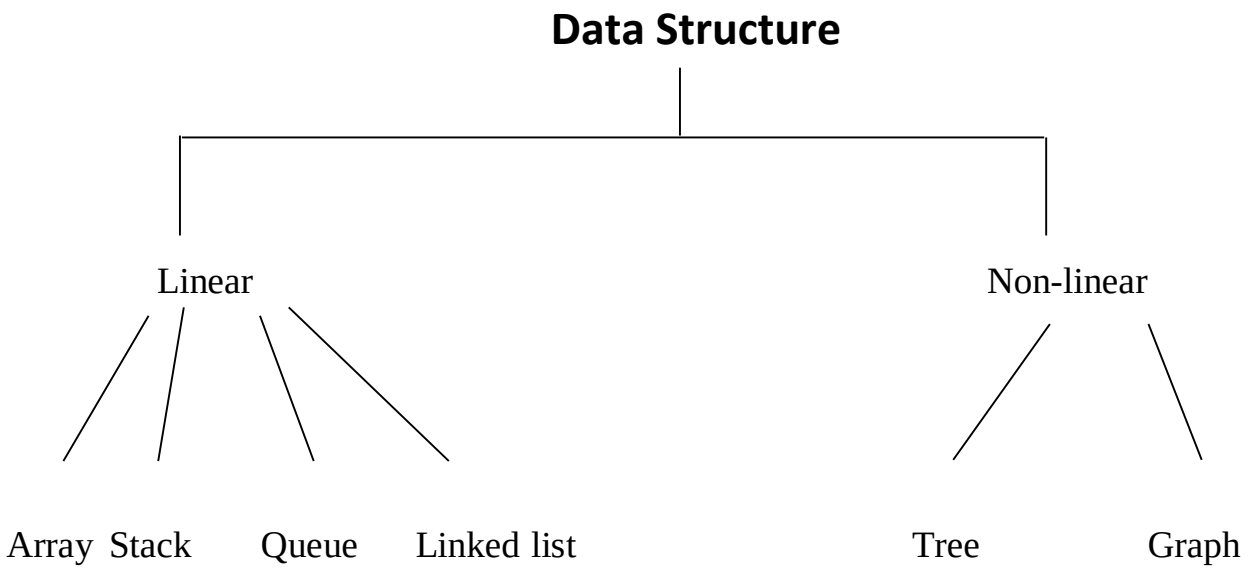
Data Structure

Data may be organized in many different ways. The logical or mathematical model of a particular organization of data is called Data Structure.

Data structure is a way of organizing and storing data so as to enable efficient access and modification of data. Different types of data structures are present and different data structures have different and unique features.

Classification of Data Structure



A data structure is said to be linear if its elements form a sequence or, in other words, a linear list. In a non-linear data structure, the elements do not form a sequence.

An **algorithm** is a finite sequence of instructions to be followed for solving a problem.

Characteristics of an Algorithm

Finiteness: An algorithm should have finite number of steps and it should terminate after a finite time.

Input: An algorithm may have many inputs or no inputs at all.

Output: An algorithm must produce at least one output as the result.

Definiteness: The steps to be performed in the algorithm must be clear, precise and unambiguous.

Effectiveness: Every instruction must be basic enough to be carried out by using pen and paper.

Array

~~The operatio~~ An array is a list of finite number of homogenous elements stored in adjacent (contiguous) memory locations. Here, "finite" means that there are specific number of elements in the array; "homogenous" means that all the elements in ~~the~~ an array are of the same type.

Sparse Matrix And Its Representation

If a lot of elements from a matrix have a value 0 then the matrix is known as a sparse matrix. If majority of elements of the matrix are 0 then an alternative through which we can store only the non-zero elements and keep intact the functionality of the matrix can save a lot of memory space.

A common way of representing non-zero elements of a sparse matrix is the 3-tuple forms. In this form, each non-zero element is stored in a row, with the 1st & 2nd element of this row containing the row and column in which the element is present in the original matrix. The 3rd element in this row stores the actual value of the non-zero element.

Examyle:- Suppose we have the sparse matrix as shown below:

$$\begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & 0 & 4 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 9 \\ 2 & 1 & 0 & 0 & 0 & 2 \\ 3 & 0 & 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 5 & 0 & 0 \end{array}$$

Its Array Representation

```
int arr[6][3] = {
    0, 1, 4,
    0, 4, 1,
    1, 4, 9,
    2, 0, 1,
    2, 4, 2,
    4, 2, 5
}
```

← The data 4 is present in row 0's and column 1's intersection position.